Aditya Kumar

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Education

University of Michigan December 2027
B.S.E. Computer Science – College of Engineering GPA: 3.5/4

Experience

Machine Learning Team Lead – University of Michigan August 2025 – Present

- Engineered multi-branch CNN pipelines (**3 architectures, 2 branches, 3 embedded convolutional blocks**) to predict cognitive load, using **NumPy** to structure and store training data and elevating deep and classical learning baseline accuracies from **78%/81%** to **87%/90%** respectively through custom protocol design
- Scaled action-unit extraction from **AWS server CPUs** to **Great Lakes GPU cluster**, cutting per-sample processing time from **30 minutes to 2–3 minutes** (up to **15x** speedup)
- Assisted subteam in training optical-flow pipeline on **Great Lakes GPU cluster** to predict heart rate, reducing prediction error from **15–20% to 6–8%**
- Currently developing multimodal deep learning architecture for automated pain assessment, integrating depth, audio, action units, and rPPG signals

ChatBot Developer – Imagine Software Jan 2025 – Aug 2025

- Built and deployed chatbot integrating **Ollama API** with **ReactJS** front-end and **Node.js** backend to handle user queries in real time
- Implemented backend PDF parsing using `pdf-parse` library to extract document data and return structured responses to front-end, enabling users to query document content directly through chatbot

Projects

Library Shelf-Order Validator – Project (GitHub) July 2026 – Present

- Built web application with **Flask** backend to validate whether library call codes are shelved in order via manual code entry, implementing naive linear-scan reference check and brute-force Longest Increasing Subsequence (LIS) algorithm to detect out-of-order entries
- Benchmarked algorithmic efficiency between linear-scan reference checks and LIS detection, finding linear-scan runs in roughly **25%** of brute-force $\mathbf{O(n^2)}$ LIS runtime; currently implementing patience sorting to optimize LIS complexity to $\mathbf{O(n \log n)}$

GamPL Developer – Hackathon September 2025

- Preprocessed IPL player dataset of **100+** matches utilizing Pandas; categorized statistical features across **5** key metrics to clean training data
- Trained RandomForestClassifier model on IPL player statistics to predict match outcomes with **76% accuracy**; hosted model as AI Agent on Fetch.AI

Involvement

Tutor – Learn To Be September 2024 – July 2025

- Mentored two students to assist development of mathematical skills, tailoring instructional strategies to meet individual student needs and support test score growth

Skills

Languages:	Java, C++, Python, JavaScript, SQL, Verilog, HTML/CSS
Frameworks & Tools:	ReactJS, Node.js, Flask, scikit-learn, Pandas, NumPy, OpenCV, AWS, Git, Ollama API
Relevant Coursework:	Data Structures and Algorithms, Computer Architecture and Security, Digital Logic Design, Linear Algebra, Calculus